Modeling how macroeconomic shocks affect regional employment: Analyzing Brazilian Formal Labor Market using Global VAR Approach

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Abstract

Assessing linkages across different regions and how macroeconomic shocks spreads out across regions is not an easy task. In this paper we address this problem using a GVAR methodology that deals with the curse of dimensionality in a ingenious form. Focusing on the Brazilian labor market, we are able to identify and quantify how a shock in aggregate economic activity spreads out regionally and throughout the time. Another novelty of our work is the use of the information collected by Brazilian Bureau of Geography and Statistics to measure how regions are linked by analyzing infrastructure linkages of Brazilian municipalities in terms of airports, roads, ports, education, health and tourism activities. Interdependence among regions is not measured only by closeness but takes in two account economic linkages. In terms of regional sensibility to macroeconomic shocks, we provide evidence these shocks tend to cause stronger effects on South, South-East and Midwest regions than Northeast and North regions. This conclusion is inline with the ideas that formal labor market is better developed in former regions than latter. South, South-East and Midwest regions in Brazil are those that have better economic and social indicators.

JEL codes: R11, E27, C23.

Key words: GVAR, cointegration, regional dynamics and labor market.

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Abstract

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1 Introduction

Macroeconomic shocks affects sectors and regions in a different intensity and delay. Population and economic activities tend to cluster in specific areas due economies of scale and agglomeration. Population tend to be near sea coast compared to interior continental areas. Sector activities tend to be concentrated in some areas or cities. Existence of a good economic infrastructure such roads, airports, ports, telecommunications facilities links region economically and are channels working on to spread out economic shocks among regions. Macroeconomic models tend to disregard regional and spatial economic activity issues due to either their aggregate nature bias or the complexity of the theme. But for a specific region it is important to disentangle how economic activity variance can be decomposed into macroeconomic and idiosyncratic factors.

In the last years time series data at disaggregated and regional levels are becoming available. At the same time an effort in the econometric literature has been done to model high dimensional problems. Two challenges must be faced to address this questions. The first is the curse of dimensionality. The second one is to model how economic regions are linked.

According to Chudik and Pesaran [2011], two approaches have been suggested to handle the curse of dimensionality: (i) shrinkage of the parameter space and (ii) shrinkage of the data. The authors deals with the curse of dimensionality by shrinking the parameter space as the number of endogenous variables tends to infinity. In this configuration, the infinite-dimensional vector autoregressive could be well approximated by a set of finite-dimensional small-scale models, and can be estimated separately in a consistent manner, which is the global vector autoregressive (GVAR) model, proposed by Pesaran et al. [2004].

Therefore, this paper proposes to use a novel approach for dealing with the curse of dimensionality when modeling the labor market. To use the GVAR methodology to model the labor market at a regional level, accounting for interdependence among regions as well as among global macroeconomic variables. The GVAR methodology eliminate the curse of dimensionality and provides a parsimonious model for labor market, and since the all variables are treated as endogenous, the GVAR allows for the global variables to be affected by the regional variables through a feedback channel.

We use information collected Brazilian Institute of Geography and Statistics that maps infrastructure and economic linkages across all municipalities in Brazil to construct the weights used in GVAR to model interdependence across regions. We do not restrict our analysis to neighborhood effects on economic activity by constructing a measure of economic connections.

Focusing on the Brazilian labor market, the aim of paper is twofold. Firstly, we want to model the link between an aggregate macroeconomic variable and regional labor dynamics and accounts for interdependence among Brazilian regions. In order to achieve that, a Global Vector Autoregressive methodology proposed by Pesaran and co-authors is used. Secondly, we want of map how regional formal labor employment reacts to Brazilian national economic activity. We employ monthly records of admissions and discharges in Brazilian formal regional labor market. The period of the study goes from 2004 to 2016 due to data constraints. The analysis of the model lead to the conclusion that the Brazilian Northeast and South region are the most elastic ones whereas the North and Central region are the less sensitive to macroeconomic shocks. Southeast region - the most developed in the country - is in the middle level.

This study is divided into sections as follows: section 2 is a brief literature review regarding previous works related to the purpose of this paper; section 3 presents the methodology used in the study; section 4 defines the models used in the paper; section 5 describes the data used with a brief description of sources and characteristics; section 6 reports the results of the study; and section 7 contains the final considerations.

2 Literature Review

This section presents a brief literature review on the literature that deals with spatial time series models and labor market.

Firstly, focusing on the Brazilian labor market, Oliveira and Carneiro [2001] analyzed the fluctuations in the employment level of several Brazilian states compared to national employment. The purpose of their study was to verify whether it is possible to establish a long-term relationship between state employment and nationwide employment. To achieve this the authors applied a cointegration analysis, a methodology proposed by Engle and Granger [1987], as well as the model of unrestricted error correction as proposed by Pesaran et al. [1996].

The authors used a database that included a comprehensive coverage of the labor market in the Brazilian states, using data from the Annual Report on Social Information (RAIS)[1] from 1985 through 1996. The results obtained from an Vector Error Correction model were consistent with those obtained with Engle and Granger method (Engle and Granger [1987]). The results lead to the conclusion that employment fluctuations in most states shares a common trend with national employment level.

The hypothesis that the formal labor market presents a different dynamics in Brazilian metropolitan and non-metropolitan areas was tested by Kretzmann and da Cunha [2009]. The goal of their study was to analyze the fluctuations in the labor market of metropolitan and non-metropolitan Brazilian areas. They used employment and unemployment data from the Ministry of Labor and Employment through the General Records of Employed and Unemployed Persons - CAGED[2] - using monthly data from 1996 through 2007. Kretzmann and da Cunha [2009] used cointegration analysis through the traditional method proposed by Engle-Granger, as well as the method proposed by Pesaran et al. [1996]. The two methods produced the similar results, suggesting that the metropolitan and non-metropolitan areas are not co-integrated.

None of these paper deals with the curse of dimensionality. The authors opt to with pairs of series. The evidence that states and national share a common trend highlight a possible channel linking macro and regional dynamics.

[1] RAIS from the Portuguese Relação Anual de Informações Sociais.
[2] Initials comes from "*Cadastro Geral de Empregados e Desempregados*" in Portuguese)

Evidence that metropolitan and non metropolitan areas do not share a common trend may be explained by the fact that neighborhood effects are not that relevant but when corrected by infrastructure connections they may be. This hypothesis is investigated in this work.

Secondly, Pesaran et al. [2004] introduced the Global Vector Autoregressive (GVAR) methodology by means of a study that modeled 11 different regions using data from 1979 through 1999. This methodology was improved by Dees et al. [2007] with the collaboration of the European Central Bank.

One of the first studies that applied GVAR to tackle interdependence across regions is Schanne [2015]. The author analyzed the regional labor market in Germany with a focus on forecasting regional labor markets. His study had two goals: the first one is to estimate a the GVAR model, as proposed by Pesaran et al. [2004]. Second, to use the developed GVAR model to construct a forecast for German employment accounting for spatial effects.

The GVAR model allows both the analysis of strong cross-section dependence and weak cross-section dependency. Consequently, Schanne [2015] concludes that the data indicate semi-strong cross-section dependence and has consistent results since Germany is a polycentric economy, in contrast with the United Kingdom or France (places where there are clearly dominant regions). The final model used in his study is a basic GVAR specification, which is a model in first differences without imposing cointegration relations, and did not include a dominant region.

In his forecasting experiment, Schanne [2015] estimated the basic GVAR model and GVAR models with leading indicators. The author used these models as a predictive forecasting tool for three and twelve months ahead – forecasting the level of unemployment in the regional divisions according to the federal employment agency of Germany. For comparison purposes, each indicator was estimated by a VAR in differences and a VAR in level. The accuracy of the prediction was evaluated by comparing the absolute mean of prediction errors (percentage) and by the Diebold-Mariano test (Diebold and Mariano [1995]). Finally, Schanne [2015] concluded that there is significant improvement in the accuracy of predictions when considering the interdependence between regions, emphasizing that the degree of improvement depends on the variables and horizon.

3 Methodology

This section presents and describes the global vector autoregressive methodology, which is the main methodology used on this paper. It is assumed that the reader is familiarized with the vector autoregressive (VAR) methodology as well as the vector error correction model (VECM) and its terminology.

Chudik and Pesaran [2011]Chudik and Pesaran (2011), considered the curse of dimensionality in the case of large linear dynamic systems. To deal with the curse they proposed to shrink the parameter space in the limit as the number of endogenous variables tends to infinity. They show that by shrinking the parameter space the infinite-dimensional VAR could be well approximated by a set of finite-dimensional small-scale models that can be consistently estimated separately, which is in the spirit of Global VAR models proposed in Pesaran et al. [2004].

Therefore, Chudik and Pesaran [2011] concludes that the GVAR approach can be used as an approximation to an IVAR featuring all macroeconomic variables. This is true for stationary models as well as for systems with variables integrated of order one.

The GVAR model is a compact model that combines individual vector error correcting models in which domestic variables are related to individual-specific foreign variables in a consistent manner.

3.1 Global Vector Autoregressive Model

The global vector autoregressive (GVAR) methodology was originally proposed by Pesaran et al. [2004] as a practical approach to construct a coherent global model. In the traditional Vector Autoregressive (VAR) methodology, an unrestricted VAR(p) model with k endogenous variables covering N countries has a large number of unknown parameters of the order $(k \cdot N)^2 p + kN$ in the mean plus $\frac{(k \cdot N+1)kN}{2}$ in the variance requiring thus a parsimonious solution. Therefore, the authors proposed the use of a model where firstly the error correction models specific to each region is estimated, with the domestic macroeconomic variables related to the corresponding foreign variables; secondly the models from each region are then combined to generate forecasts for all variables. In short, the GVAR model combines VECM models of each region, so that domestic variables are related to regionally specific foreign variables in a consistent manner. Therefore, the model is able to take into account the interdependencies between economic regions in the analysis.

Subsequently Dees et al. [2007] improved the GVAR model with the cooperation of the European Central Bank. This version of the GVAR model included 26 countries (covering 90% of world production), with the euro area being considered as one sole economy with estimates from 1979 to 2003. The version of the GVAR model on this paper will be presented in this study is based on that of Dees et al. [2007].

The GVAR model can be summarized as a two-step procedure. In the first stage, specific models of each region are estimated conditioned to external influences of the other regions. That is, the models are represented as VAR models, which have domestic variables and foreign variables, the latter being treated as weakly exogenous. [3] In the second step, the VAR models of each region are grouped into one single global VAR model. The GVAR model, as well as any VAR model, can be used for analyzing shocks in the economy, and predictions.

3.1.1 Specific models for each region

For the specific models used in the first stage of the GVAR methodology, consider a panel with N elements, each element showing k_i variables observed during periods $t = 1, 2, \cdots, T$. Within this panel structure the following variables are defined: $x_{i,t}$ as a vector $k_i \times 1$ of domestic variables for the element i in period t; x^*_{it} as a vector $k^* \times 1$ of foreign variables; and $u_{i,t}$ as an error vector. Thus, following the notation of Chudik and Pesaran [2016], the specific models of each element consist of a set of domestic and foreign variables. Specifically, for any i element:

[3] See Engle et al. [1983] for a definition.

$$x_{i,t} = a_{i0} + a_{i1}t + \sum_{j=1}^{p_i} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x^*_{i,t} + \sum_{l=1}^{q_i} \Lambda_{il} x^*_{i,t-l} + u_{i,t} \tag{1}$$

where Φ_{ij} is an array of lag coefficients for lag j associated with domestic variables and Λ_{il} is an array of lag coefficients for lag l associated with foreign variables.

The GVAR model treats foreign variables as weighted averages of the domestic variables of each corresponding element, and the weights are specific to each unit. Therefore, the foreign variables of each element i can be computed as a weighting of the domestic variables of other units in the panel. A weighting matrix W_{ij}, where $j = 1, 2, \cdots, N$ is defined as a set of weights with the following characteristics: $W_{ii} = 0$ and $\sum_{j=1}^{N} W_{ij} = 1$. Thus, the foreign variables can be defined as:

$$x^*_{i,t} = \sum_{j=0}^{N} W_{ij} x_{jt} \tag{2}$$

The weights are predetermined and used to capture the importance of region j for the economy of region i.

From the domestic variables and the foreign variables estimated from the weighting previously mentioned, VECM models are estimated separately for each region, considering that in the VECM models the foreign variables $x^*_{i,t}$ are treated as weakly exogenous. The use of the VECM models allows both the co-integration between domestic variables, and between domestic and foreign variables.

3.1.2 Global model

Although estimation is performed individually, element-by-element, the GVAR model is solved considering all variables as endogenous to the system. Thus, from the specific model of each element, $Z_{i,t}$ is defined as:

$$Z_{i,t} = \begin{bmatrix} x_{i,t} \\ x^*_{i,t} \end{bmatrix} \tag{3}$$

Using (3) in (3) it is possible to obtain:

$$A_{i0} Z_{i,t} = a_{i0} + a_{i1}t + \sum_{l=1}^{p} A_{il} Z_{i,t-l} + \varepsilon_{i,t} \tag{4}$$

where $p = \max_i\{p, q_i\}$, $A_{i0} = [\begin{array}{cc} I_{k_i} & -\Lambda_{i0} \end{array}]$, $A_{il} = [\begin{array}{cc} \Phi_{il} & \Lambda_{il} \end{array}]$ for $l = 1, 2, \cdots, p$. And define $\Phi_{il} = 0$ for $l > p_i$, and $\Lambda_{il} = 0$ for $l > q_i$.

Let $x_t = [x'_{0,t}, x'_{1,t}, \cdots, x'_{N,t},]'$ be a vector $k \times 1$ of all the variables in the panel, where $k = \sum_{i=1}^{N} k_i$, W_i can be defined as being a matrix of dimension $(k_i + k^*_i) \times 1$ of known constants defined in terms of the individual weights W_{ij}. Therefore, the variables of the specific models are defined as a function of x_t.

$$Z_{i,t} = W_i x_t \tag{5}$$

Using this identity, the individual models can be written as:

$$A_{i0} W_i x_t = a_{i0} + a_{i1} t + \sum_{l=1}^{p} A_{il} W_i x_{t-l} + \varepsilon_{i,t} \tag{6}$$

It is possible to combine the new individual models into one single model, and this can be done by "stacking up" each model in a large-scale VAR.

$$G_0 x_t = a_0 + a_1 t + \sum_{l=1}^{p} G_l x_{t-l} + \varepsilon_t \tag{7}$$

Where:

$$G_0 = \begin{bmatrix} A_{00}W_0 \\ A_{10}W_1 \\ \vdots \\ A_{N0}W_N \end{bmatrix} \quad G_1 = \begin{bmatrix} A_{0l}W_0 \\ A_{1l}W_1 \\ \vdots \\ A_{Nl}W_N \end{bmatrix} \quad a_0 = \begin{bmatrix} a_{00} \\ a_{10} \\ \vdots \\ a_{N0} \end{bmatrix} \quad a_1 = \begin{bmatrix} a_{01} \\ a_{11} \\ \vdots \\ a_{N1} \end{bmatrix} \quad \varepsilon_t = \begin{bmatrix} \varepsilon_{0,t} \\ \varepsilon_{1,t} \\ \vdots \\ \varepsilon_{N,t} \end{bmatrix}$$

Since the G_0 matrix is a non-singular matrix, which depends only on the estimated parameters and the link matrix W_i, it is possible to rewrite the equation (7) as:

$$x_t = b_0 + b_1 t + \sum_{i=1}^{p} F_i x_{t-i} + \epsilon_t \tag{8}$$

Where:

$$\begin{aligned} F_i &= G_0^{-1} G_i \\ b_0 &= G_0^{-1} a_0 \\ b_1 &= G_0^{-1} a_1 \\ \epsilon_t &= G_0^{-1} \varepsilon_t \end{aligned}$$

Hence, the global model, which consists of a set of individual models from each region, can be solved recursively. With that it is possible to consistently determine the coefficients of the model.

3.1.3 Dominant Unit Model

Global VAR is based on the hypothesis of weak exogeneity (Engle et al. [1983]). If this hypothesis fails for one or regions, these regions must be modeled jointly. In spatial analysis it is possible that some important regions can affect the others in a specific form. This information must be included in the model. Therefore, a vector ω_t of common variables and its lagged values augment each specific model. So, each element model can be stated as:

$$x_{i,t} = a_{i0} + a_{i1}t + \sum_{j=1}^{p_i} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x^*_{i,t} + \sum_{l=1}^{q_i} \Lambda_{il} x^*_{i,t-l} + D_{i0}\omega_t + \sum_{l=1}^{s_i} \Psi_{il}\omega_{t-l} + u_{i,t} \tag{9}$$

The marginal model for the dominant variables can be estimated with the feedback effects from x_t, or not. The feedback from the variables in the GVAR model are captured via cross-section averages, this is done by augmenting the dominant unit model with lags of $x^*_{\omega t}$.

$$\omega_t = \sum_{l=1}^{p_\omega} \Phi_{\omega l}\omega_{t-l} + \sum_{l=1}^{q_\omega} \Lambda_{\omega l} x^*_{i,t-l} + \eta_{\omega t} \tag{10}$$

Different lag orders for the dominant variables (p_ω) and cross-section averages (q_ω) could be considered. It is worthy of pointing out that contemporaneous values of foreign variables do not feature in the equation.

3.2 Regional Weights

Finally, the number of connections between regions were obtained from IBGE [2008]. This work aims to determine economic linkage among municipalities in Brazil. The study not only analyzed regular transport links, such as those that go to the urban centers, but also the main destinations of the residents to obtain products and services, such as general goods, education, airport travel, and health services. The acquisition of agricultural inputs and the destination of the agricultural products were also analyzed in the study. The main result of the paper is a Brazilian urban network map that connects all 5,564 Brazilian municipalities. This urban network contains the hierarchy of the urban network and the regions of influence of each urban center.

A connection chart between all the Brazilian cities is constructed by merging the links provided in the urban network framework, which are the main links between municipalities, with the links regarding airport connections, shopping destinations, education, health services, and recreation activities.

Since every city belongs to a mesoregion the links between mesoregions are also determined with the following characteristics: if the two cities are within the same region then the link between them is an internal link; if the two cities are in different mesoregions then the link between them is a connection between the two mesoregions.

As mentioned earlier the weight matrix is a function of the number of connections between regions, however at this point it is worth mention that some connections are more important than others, however, since the measurement of importance for each connection between each region is unfeasible, an equal weight to all connections is assumed, resulting in a linear function, which is the average.

The weight matrix to a region i can then be determined by the number of connections to other regions divided by its total number of connections. Therefore, a weight vector for each region is determined as:

$$W_i = \frac{\begin{bmatrix} C_{i1} & \cdots & C_{iN} \end{bmatrix}}{\sum_{j=1}^{N} C_{ij}} \tag{11}$$

where C_{ij} is the number of connections that originated at region i and has region j as destination, keeping in mind that connections within the same region ($i = j$), also called internal links, have a zero weight ($C_{ii} = 0$).

One of the most important steps in GVAR methodology is the construction of the weight matrix. In Pesaran's initial work, the weight matrix is defined by country-specific trade weights. In fact, the weight matrix W_i is designed to capture the importance of all other regions for the region i (Di Mauro and Pesaran [2013]). So, to determine the weight matrix it is necessary to determine the importance of each region in respect to every other region.

Focusing on the labor market, to determine the importance of a region to another is to determine the importance of one labor market to another labor market. Each region has its own labor market with firms and workers, and since firms can't change markets it is the workers that freely change markets. In other words, a worker can look for a job in other regions, but a firm must attract the workers from other regions to their own market. Consequently, for a worker to commute to another region there must be a connection between these regions.

The connections between regions goes beyond neighborhood effects in an integrated economy. There are many others channels such as physical infrastructure that allow workers to commute (by car, by train, by plane, etc.) to another region. Therefore, regions with a higher connections are more important than regions with fewer connections to model a specific unit.

Consequently, the importance of region j to region i will be a function of the connections between them, and the weight matrix will be a based on the number of connections between the regions.

4 Database Description and Model Specification

The labor market interacts with the macroeconomic environment in many ways. One of these channels is the level of economic activity. Industrial production is used as a proxy to economic activity. It is highly correlated to GDP but available at a higher frequency level. The first is available quarterly whereas the latter is monthly.

Finally, an attempt to model the Brazilian labor market at municipality was done but discarded. Brazil had a total of 5564 municipalities in 2005 according to Brazilian Geography and Statistics Bureau. This government institute grouped municipalities into 137 mesoregions, and 27 states. Throughout many municipalities were merged or split and it is difficult to construct long time series. Therefore we opt to work at meso-regional level as by Brazilian Institute of Geography and Statistics in 2015.

4.1 Database

Table 1 contains a detailed description of data source The following section is a brief description of the data used in this paper, as well as its sources, frequency.

Data	Source	Frequency	Period
Brazilian Industrial production	IBGE (1)	Monthly	2004-01 to 2016-12
Employment and Unemployment	CAGED (2)	Monthly	2004-01 to 2016-12
Mesoregions and municipalities	IBGE (1)	N/A	as reported on 2015
Connection between regions	IBGE (1)	N/A	as of 2007

Source: elaborated by the author
(1) Brazilian Institute of Geography and Statistics
(2) Ministry of Labor and Employment through the General Records of Employed and Unemployed Persons.

Table 1: Data sources and characteristics

The data for the Brazilian Industrial production was obtained from IBGE. The employment and unemployment records, which are the workers admitted to and discharged from the labor market, was obtained from the Ministry of Labor and Employment through the General Records of Employed and Unemployed Persons. The data was obtained at the municipality level and grouped by mesoregions. The frequency of the data is monthly and goes January 2004 to December 2016.

The mesoregions definitions, as well as the information of total population, microregions, mesoregions, state, and their designation codes, were obtained from the Brazilian Institute of Geography and Statistics (IBGE from the Portuguese Instituto Brasileiro de Geografia e Estatística) as reported on the year 2015.

4.2 Descriptive statistics

Table 2 shows the descriptive statistics for the industrial production as well as the admitted and disconnected series for the mesoregions that contain the cities of São Paulo, Rio de Janeiro and Brasilia - three of the most important cities in Brazil. The descriptive statistics for all the regions can be found in the appendices.

All models were estimated using MatLab software R2016b and the GVAR Toolbox 2.0. The following MatLab add-ons were installed during the estimation of the models: Mapping Toolbox version 4.4; Econometrics Toolbox version 3.5; Optimization Toolbox version 7.5; Statistics and Machine Learning Toolbox version 11.0.

The GVAR Toolbox 2.0, developed by Smith and Galesi (2014), is and open source collection of MatLab procedures with a Microsoft Excel based interface, designed for GVAR modeling purposes. It's worth mentioning that the GVAR toolbox does not require any MatLab add-on.

Region	Series Name	Mean	Median	Max	Min	Std. dev.	Skewness	Kurtosis	Jarque-Bera
Global	Ind. Prod.	4.5503	4.5502	4.7238	4.3041	0.0963	-0.3792	2.59	4.7326
Sp - Metropol Sao Paulo	Admitted	226073	223610.5	329823	115254	55443.5	-0.0856	1.7569	9.8751
Sp - Metropol Sao Paulo	Discharged	212768.2	217592	298117	107413	56594.5	-0.307	1.8034	11.4459
Rj - Metropol do RJ	Admitted	95295.5	94490	140206	56900	22461.6	0.1119	1.7417	10.2581
Rj - Metropol do RJ	Discharged	90694.9	92642.5	141759	51785	23537.8	-0.0474	1.7966	9.1128
Go - Distrito Federal	Admitted	22942.6	22685.5	36420	12557	5643.1	0.0728	1.738	10.1272
Go - Distrito Federal	Discharged	22068.6	22648.5	33552	10988	5931.7	-0.1269	1.7852	9.6577

Source: elaborated by the author

Note: The data for the industrial production is presented at log level.

Table 2: Descriptive statistics of the variables

4.3 GVAR specification

The objective of this study is to use the GVAR methodology, as described in the previous section to assess the Brazilian labor market model. Each Brazilian mesoregion will be a GVAR unit.

Since the labor market model will consist of workers admitted to and discharges from the labor market, the specific model for each mesoregion will contain the information of those admitted to and disconnected from the labor market in the period t for that region. Therefore, for each mesoregion i a vector x_{it} can be defined as:

$$x_{i,t} = \begin{bmatrix} Adm_{i,t} \\ Dis_{i,t} \end{bmatrix} \tag{12}$$

where $Adm_{i,t}$ is the number of workers admitted (employment flow) to the labor market in period t in region i; $Dis_{i,t}$ is the number of workers disconnected (unemployment flow) from the labor market in period t in region i.

Each region experiences an influence from other regions. Using the GVAR methodology the foreign variables are defined as a vector $x^*_{i,t}$ as follows:

$$Adm^*_{i,t} = \sum_{j=1}^{N} W_{ij} Adm_{j,t} \tag{13}$$

$$Dis^*_{i,t} = \sum_{j=1}^{N} W_{ij} Dis_{j,t} \tag{14}$$

$$x^*_{i,t} = \begin{bmatrix} Adm^*_{i,t} \\ Dis^*_{i,t} \end{bmatrix} \tag{15}$$

where: W_{ij} is a link matrix that captures how much the region j influences region i; $Adm^*_{i,t}$ is the influence in region i of the number of workers admitted to the labor market in other regions in period t; $Dis^*_{i,t}$ is the influence in region i of the number of workers disconnected from the labor market in other regions in period t;

The specific models are defined with unrestricted intercepts and no trend coefficients. It is worth-mention that even though a deterministic trend is not included in the model, the levels of the domestic variables may be trended due to the drift coefficients.

$$x_{i,t} = a_{i0} + \sum_{j=1}^{k_{1,i}} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x^*_{i,t} + \sum_{l=1}^{k_{2,i}} \Lambda_{il} x^*_{i,t-l} + u_{i,t} \tag{16}$$

where $k_{1,i}$ and $k_{2,i}$ are the lag order for the domestic variables and foreign variables respectively. The model is defined with a maximum 2-lag order for domestic and foreign variables, where the lag selection for each region will be done by the GVAR toolbox using Akaike Information Criterion.

4.4 Macroeconomic Dynamics

In the GVAR model macroeconomic variables are treated as common variables and modeled in a Dominant Unit. Therefore, the Dominant Unit of the GVAR model will contain the log level of the Brazilian industrial production defined by the vector ω_t, with a feedback variable from the regions through the aggregate values of the employment and unemployment. So, the Dominant Unit will be a univariate model defined in first differences with a trend included in the levels, and can be expressed as:

$$\omega_t = \begin{bmatrix} \log(IndProd_t) \end{bmatrix} \tag{17}$$

$$\Delta\omega_t = \Phi_{\omega 0} + \sum_{l=1}^{k_1} \Phi_{\omega l} \Delta\omega_{t-l} + \sum_{l=1}^{k_2} \Lambda_{\omega l} \Delta x^*_{i,t-l} + \eta_{\omega t} \tag{18}$$

where Δx^* is the feedback variable containing the aggregate values of the employment and unemployment from all regions; k_1 and k_2 the lag order for the global variable (Industrial Production) and feedback variables respectively. The maximum lag order is defined as 3 for both, the global variable and the feedback variables, where the lag selection for each variable will be set by the GVAR toolbox using Akaike Information Criterion.

5 Results

Modeling a system such as the Brazilian labor market at the mesoregions level is naturally subject to considerable uncertainties. There are many choices to be made for each individual mesoregion model, for example: the variables to be included in the specific models; the lag order of the domestic variables; the lag order of the foreign variables; the number of cointegrating relations; and whether to impose long-run and short-run restrictions on the parameters are just a few of the choices.

As explained earlier the Global VAR model comprises several specific models, solved in a two-step process. The model is defined using the data for admissions and discharges for every mesoregion as well as the weight matrix defined previously. All regions were set to be affected by foreign variable and the dominant unit.

The final model was estimated using the GVAR automatic lag selection with the Akaike information criterion and a maximum lag order of 2 for domestic as well as foreign variables. A total of 137 specific models were specified. Lags for domestic variables were set to 2 in 98 models, and set to 1 in the other 39 models. The lag for foreign variables were set to 2 in 91 models, and set to 1 in the other 46 models.

Existence of cointegration among variables were investigated using automated procedure of GVAR toolbox 2.0. All individual models had the deterministic components treated as an unrestricted intercept and no trend in the error correction models. Out of the 137 estimated models, 9 were specified with 1 cointegration relationship, and 128 models were specified with 2 cointegration relationships.

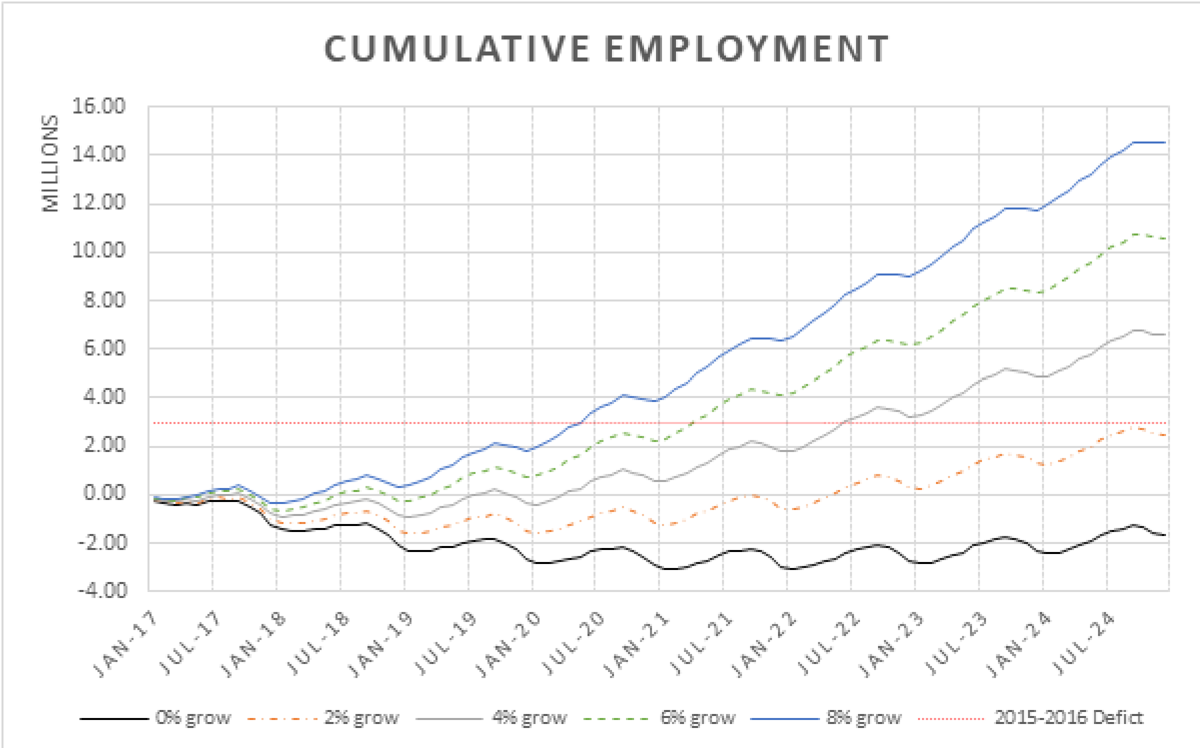
The GVAR model determines a model for each region considering foreign effects, as well as effects from the variables in the dominant unit model. With no overidentifying restrictions imposed, the final model proved to be stable with 815 eigenvalues lying inside the unit circle 10 of which lying on the unit circle.

5.1 Forecast Exercise

Using the GVAR methodology, it is possible to estimate dynamics response of unemployment and employment in each region considering not only the industrial production but also the interdependence of each region. With the use of conditional forecasts it is possible to evaluate the behavior of the labor market in response to a change in the global economy, in our case represented by the industrial production. Therefore a series of conditional forecasts for different growth scenario of industrial production were estimated to assess the impact of the industrial production on the labor market. The series of conditional forecasts of the labor market will consist of a base scenario and several other growth scenarios for the Brazilian industrial production.

The baseline scenario for the industrial production growth will be a 0% growth coefficient over the last year in the series, in this case the year of 2016. The other growth scenarios on the set will be: 2% growth coefficient; 4% growth coefficient; 6% growth coefficient; and 8% growth coefficient over the last year.

The industrial production growth is made by applying the annual growth coefficient in each month compared



Source: elaborated by the author

Note: 2015-2016 Deficit: The net employment deficit accumulated from 2015 through 2016

Figure 1: Brazilian accumulated employment forecast for different growth scenarios

to the last year. In other words, a 2% growth coefficient means a 2% grow in January 2016 compared to January 2017 and so on. All the scenarios were analyzed over an eight-year forecast.

5.1.1 Forecast, net employment and accumulated employment

Net employment gain will be defined as the difference between admissions minus discharges. Also “total net employment mesoregion gain” is defined as the sum of the net employment gain for every municipality of the mesoregion, and consequently starting the forecast exercise started in January 2017, the cumulative employment gain is the sum of all the previous “total net employment” until that year. The total cumulative employment is an aggregate of all mesoregions and can be used to evaluate how the Brazilian labor market will behave in different macroeconomic scenarios in terms of changes in stocks of employment.

From 2015 through 2016 the total net employment gain is negative of minus three millions due to the fact that Brazilian economy was facing a deep recession. Taking December 2016 as reference, the baseline forecasts reveal that a 0% annual growth for industrial production would imply a great loss of formal employment. If industrial production starts to recover at rate of 2% annual growth then the net gain will become positive only about September of 2021. At end of forecast horizon period a cumulative gain of 2.5 million net jobs will be created. A 4% annual growth in the industrial production would cause a significant improvement to cumulative net employment gain with the first positive net gain of the series occurring in September 2017 and an cumulative gain of employment will be

Source: elaborated by the author.

Figure 2: Accumulated employment rate variation for various industrial production growth scenarios.

6.5 million by the end of 2024. For a 6% and 8% annual growth cumulative gain will become positive in June 2017, and will end forecast periods with a value of 10.5 million and 14.5 million, respectively. It is worth mention that a 4% annual growth in the industrial production the Brazilian cumulative net employment deficit from 2015 to 2016 would compensated only in July 2022.

5.2 Industrial production elasticity by region

The net employment defined in the last section can be interpreted as the variation on the total employment level. In order to normalized the valuer we divide each this variable by total population of each meso-region. This gives an idea about the dynamics of each region. It would be better to have an estimate of the population at labor force force but this is not available. Therefore, chnages on employment rate that will be used in this paper is as a proxy employment rate creation.

Since the GVAR methodology allows for a forecast for each region, it is possible to map changes of the employment rate for every region, where the cumulative changes is the sum of all previously variations. Therefore, it is possible to evaluate how a change in the industrial production affect the labor market in each region, in other words it is possible to establish the employment elasticity in respect to the industrial production.

Data on each mesoregions are grouped into five Brazilian regions (North, Northeast, Southeast, South, Central).

Taking the IBGE estimated population for 2016 as the total population, and 0% growth as a base scenario, it is possible to obtain an industrial production elasticity of the employment for each major region over an eight-year forecast. Considering a 2% annual growth on the industrial production, the South region is the most sensitive with 0.18 elasticity, the second most sensitive region is the Northeast with a 0.15 elasticity. The Southeast region has a 0.10 elasticity, while the least sensitive major regions are the North and Central, each one with 0.06 elasticity.

This indicates that the Northeast and South regions are those with higher elasticity, being strongly affected by changes in industrial production. The Southeast region, which is the most developed, has an intermediate elasticity. This is probably due to the high development level of the region, making its marginal gain not as high as in the Northeast and South regions.

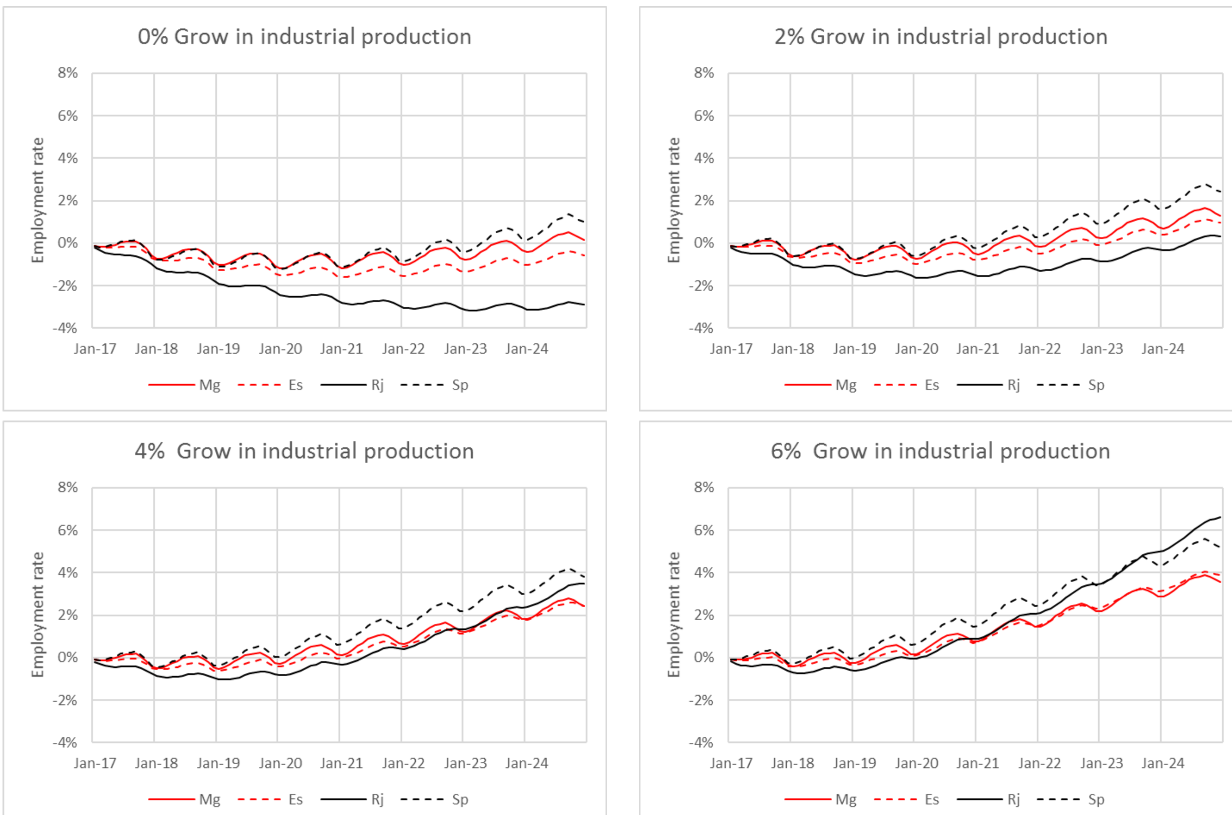
Although previous analysis was made with macro regions, GVAR methodology allows to make the same analysis in a more in-depth level. Since the GVAR equations are solved for every region, we can evaluate not only the state elasticity but also the elasticity for every mesoregion.

Figure 3 shows the estimated employment rate change for the Southeast region, the most developed in the country. Using the 0% growth as a base scenario the elasticity for the states of Minas Gerais, Espírito Santo, Rio de Janeiro and São Paulo are estimated. Again, considering a 2% annual growth on the industrial production, Rio de Janeiro is the most sensitive state with 0.19 elasticity. Followed by Espírito Santo with a 0.09 elasticity. Minas Gerais and, São Paulo are estimated to have an elasticity of 0.07 and 0.08, respectively.

As commented earlier with the GVAR model it is possible to evaluate every region in a more in-depth analysis. Since listing all 137 regions estimated elasticity would be impractical, the region of São Paulo metropolitan area (one of the most important in the country) is compared along with 4 other randomly selected regions, which are: Goiás Central region, Borborema region, Jequitinhonha region, and Porto Alegre metropolitan area. Goiás Central region is the most populous, rich and densely populated mesoregion of the Goiás state, with 3.1 million inhabitants and a GDP per capita estimated at 11.8 thousand Brazilian Reais. The mesoregion of Borborema is a dry land, with approximately 310 thousand inhabitants, its GDP per capita is estimated to be 5 thousand Brazilian Reais. The Jequitinhonha Valley is a region known due to its low social indicators. It has an GBP per capita of 2.6 thousand Brazilian Reais and 731 thousand inhabitants. Porto Alegre metropolitan area an important region in the South, it's economy is strongly influenced by the industrial sector, with a 12.1 thousand Brazilian Reais GDP per capita and 5 million inhabitants.

Figure 4 shows the estimated elasticity for the regions. As previously, the 0% growth is used as a base scenario the elasticity. With a 2% annual growth scenario on the industrial production, Porto Alegre is the most sensitive mesoregion with 0.19 elasticity. Followed by São Paulo with a 0.16 elasticity. The Goiás central area has a moderate elasticity with a value of 0.07. Lastly the Borborema and Jequitinhonha mesoregions proven to be very inelastic, with only 0.02 elasticity.

The most populated mesoregions in Brazil are São Paulo metropolitan area, Rio de Janeiro metropolitan area,



Source: elaborated by the author

Mg: Minas Gerais; Es: Espírito Santo; Rj: Rio de Janeiro; Sp: São Paulo.

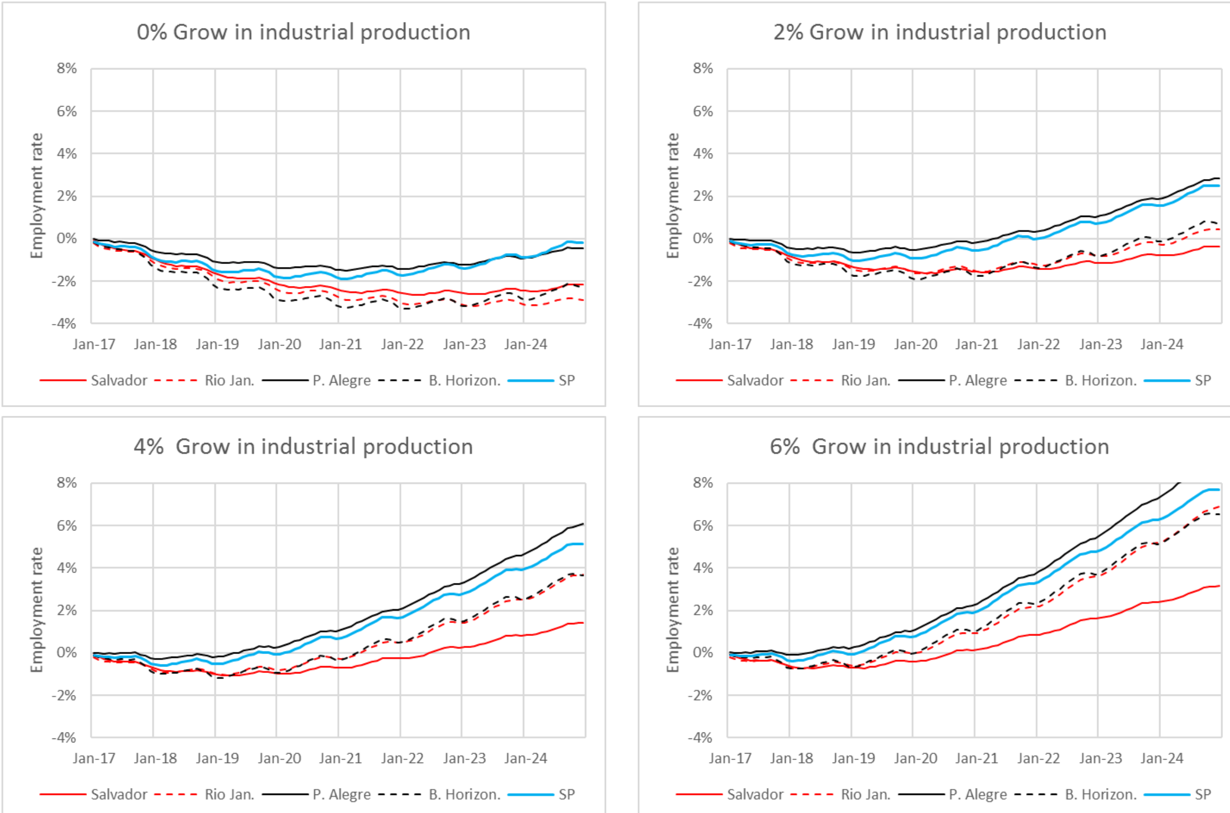
Figure 3: Accumulated employment rate variation for various industrial production growth scenarios on state level.

Source: elaborated by the author

C.Go: Goiás central area; Borb: Borborema; Jequit: Jequitinhonha;

P.Alegre: Porto Alegre metropolitan area; SP: São Paulo metropolitan area

Figure 4: Cumulative employment rate changes for various industrial production growth scenarios on mesoregion level.



Source: elaborated by the author

Salvador: Salvador metropolitan area; Rio Jan: Rio de Janeiro metropolitan area; P.Alegre: Porto Alegre metropolitan area; B.Horizon.: Belo Horizonte metropolitan area SP: São Paulo metropolitan area

Figure 5: Accumulated employment rate variation for various industrial production growth scenarios on mesoregion level – Most populated mesoregions.

Belo Horizonte metropolitan area, Porto Alegre metropolitan area, and Salvador metropolitan area. Rio de Janeiro metropolitan area has approximately 12.9 million inhabitants. Belo Horizonte metropolitan area in its turn has approximately 6.7 million inhabitants. Salvador metropolitan area has around 4.7 million inhabitants according to estimate from Brazilian Geography and Statistic Bureau in 2017.

Figure 5 shows estimated elasticity for the most populated mesoregions in Brazil. As before, the 0% growth is used as a base scenario the elasticity. With a 2% annual growth scenario on the industrial production, the metropolitan area of Rio de Janeiro and Porto Alegre are the most sensitive mesoregions with 0.19 elasticity. Belo Horizonte and São Paulo have an elasticity of 0.17 and 0.16 respectively. Lastly the metropolitan area of Salvador has an 0.11 elasticity.

5.3 Regional dynamics

One of the advantages of the GVAR methodology is being able to detect the dynamic interactions between the regions in the model. In other words, the GVAR model can provide a forecast with the dynamics of each region. Consequently, the final analysis of our study will be to evaluate the interactions of the regions among different scenarios.

We discuss the dynamics of employment on four scenarios for up to seven years ahead. Initial period is December of 2016. Scenarios to be compared will be the 0% growth (baseline), a 2% growth, and 4% growth in Industrial production, and the forecast horizons will be one year, three years, five years, and seven years ahead.

It is expected that the most populated areas have a higher number of admissions and discharges, or a higher net employment, than less populated areas. Therefore, to account for the total population in the region, the comparison between the regions will be made using the previously defined employment rate.

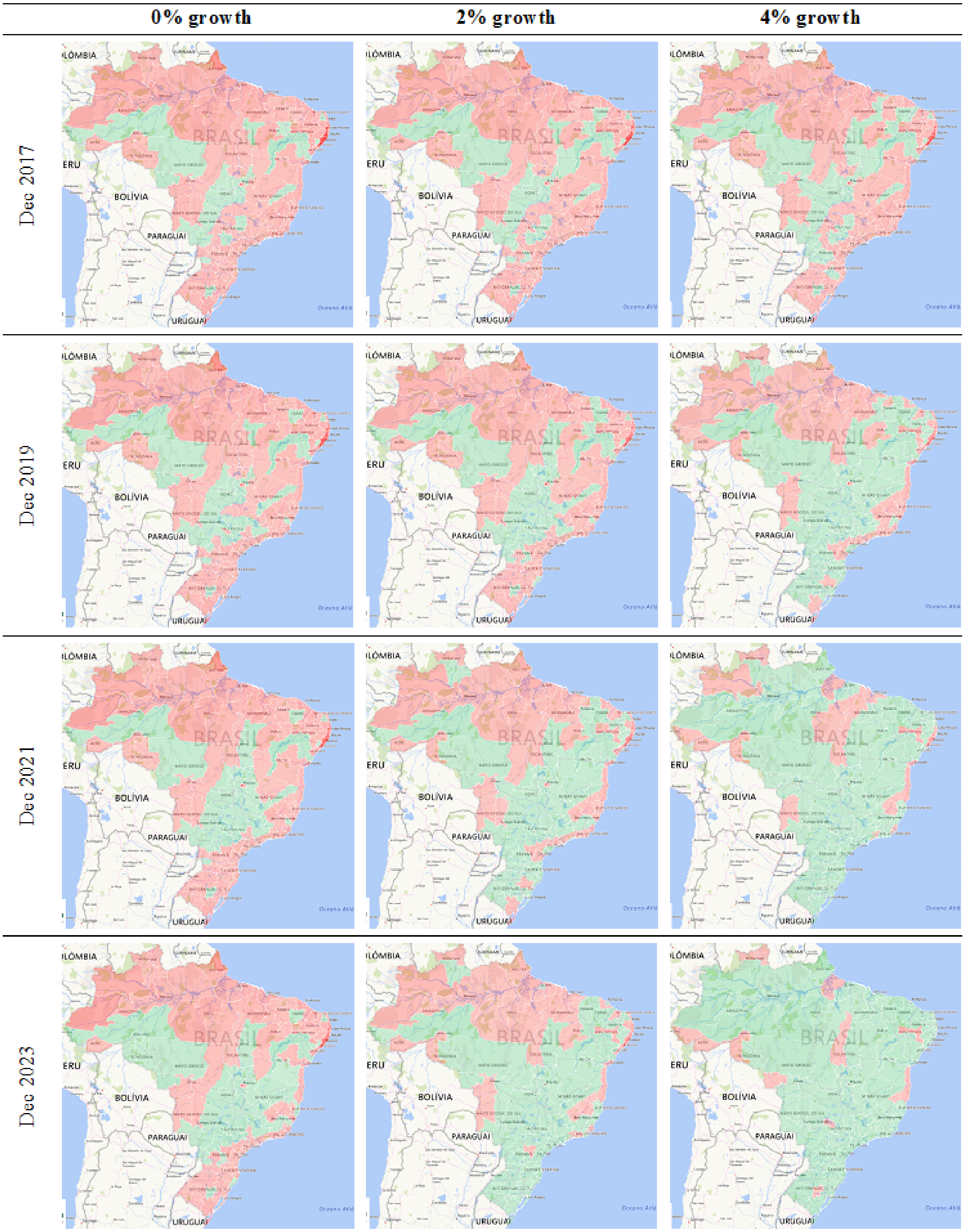
By using the GVAR methodology and comparing the accumulated employment rate it is possible not only to determine how the employment rate in each region reacts to a variation in the industrial production, but also to determine the dynamics of the interactions of the regions. The South, Southeast and Northeast regions prove to be very dynamic, displaying a constant interaction within its regions, while the North and Central regions appear to get little influence from the neighboring regions.

As shown in Figure 6, the 0% growth scenario shows almost no improvement in the individual regions. On December 2017, this scenario has 112 mesoregions, which show a negative net employment. On December 2023, only 28 regions were able to develop a positive balance of employment, and 84 mesoregions still show a negative net employment (the 84 regions are divided as: North: 15; Northeast: 33; Southeast: 16; South: 14; Central: 6). The 2% growth scenario shows some improvement over the 0% growth, with only 41 mesoregions with negative net employment on December 2023 (North: 14; Northeast: 18; Southeast: 6; South: 1; Central: 2). Finally, the 4% scenario shows a significant improvement, only 13 among all 137 mesoregion regions have a negative net employment by December 2023 (North: 6; Northeast: 3; Southeast: 2; South: 1; Central: 1).

6 Limitations and possible extensions

This work can expanded in many forms. First a more comprehensive macroeconomic model can be formulated and different types of macroeconomic shocks can be identified and their impact simulated. Secondly, it is possible to construct a "tailor-made" GVAR. Each individual model can be estimated using recent developed techniques for model selection such as Autometrics (Hendry and Doornik [2014]). This route was explored by Ericsson and Reisman [2012] but not in a regional model. It is also possible to disagregated data not only by regions but also by sectors such as agriculture, industry and services among others.

Another possible path of research consists of evaluating predictive performance at different aggregate levels and horizon against simple benchmarks such as random walk or first autoregressive.



Source: elaborated by the author.
Original data from: CAGED

Figure 6: Dynamic comparison of the accumulated employment rate between different growth scenarios

7 CONCLUSION

The initial purpose of this paper is a twofold: firstly, creating a model of the Brazilian labor market at a regional level using the GVAR methodology, as proposed by Pesaran et al. [2004]. Secondly, using the developed model to estimate the regional elasticity of employment in respect to the economic activity of the country, identifying the dynamics of the regions as well as infer on the effects of the global variables on the labor market.

The increasing integration of the Brazilian territory has generated great interdependence among its regions. This interdependence plays a key role in the modeling and analysis of the markets. Therefore, macroeconomic analyzes should consider not only the existence of a correlation between local and global variables, but also the correlation with variables from other regions through this interdependence channel.

In this scenario, the GVAR methodology, developed by Pesaran et al. [2004], plays a leading role in the modeling of the labor market, since it accounts for the interdependence and connections between economies and regions. Therefore, if the assumptions of weak exogeneity is maintained, it is possible to use the GVAR to construct a simple and practical model for the labor market.

We document existence of cointegration relationship between regions using our indicator of economic linkage that goes beyond neighborhood effects. Using GVAR model we are able to consolidate all the in a comprehensive. The model can be used to simulate evolution of employment under different macroeconomic scenarios and unveil and quantify linkages across regions.

Thus, using the previously estimated model, several forecast scenarios for the labor market are analyzed, leading to the assessment that if industrial production remains at the current level. In other words a 0% annual growth coefficient, the labor market will keep a negative employment balance, meaning that the economy will remain in recession at least for the next 8 years. On the other hand, a 2% annual growth of the industrial production shows significant improvement in the accumulated net employment leading to a positive accumulated balance of 2.5 million employment positions for an 8-year forecast. Higher growth scenarios (4%, 6% and 8% annual coefficients) also show higher accumulated employment. Therefore, an increase on the industrial production affects the labor market in a positive correlation, as expected according to the macroeconomic theory.

Focusing on Brazilian main regions -North, Northeast, Southeast, South, and Central - and using the forecasts scenarios, it was possible to estimate the elasticity of employment in relation to industrial production for each major region. The most sensitive region is the South followed by the Northeast; they have an elasticity of 0.18 and 0.16, respectively. The Southeast, the country's most developed one, has an intermediate elasticity whereas North and Central regions have the lowest ones.

Finally, results from the model suggests a rich dynamic relationship among mesoregions. The South, Southeast and Northeast are quite dynamic and presents a higher level of interaction whereas while the North and Central regions these links appears not to be as stronger.

Finally our paper suggests that GVAR methodology can give good insight on the dynamics of formal labor employment and deals with curse of dimensionality in a manageable form.

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